1. Project Summary

This project investigates the relationship between workplace mental-health culture and national suicide mortality rates. The central question asks whether supportive organisational practices such as providing mental-health benefits or encouraging open discussions are associated with lower suicide rates across countries. To examine this, two complementary datasets were used.

The first dataset, *OSMI Mental Health in Tech Survey (2016)* (osmi-survey-2016_1479139902.json), captures self-reported perceptions from more than 1 400 technology professionals worldwide regarding mental-health benefits, comfort discussing psychological issues, and perceived stigma in their workplaces. The second dataset, *World Health Organization Global Suicide Rates (2016)* (b23d20bb-3eae-46c1-b757-3eaa35e8ba86.csv), provides official national suicide-mortality figures per 100 000 people. By integrating these two data sources, one representing micro-level workplace experiences and the other macro-level public-health outcomes. The study aims to explore how corporate mental-health support may reflect or reinforce national wellbeing.

The findings are presented through a structured data-wrangling pipeline and three key research questions that analyse the correlation between mental-health benefits, openness to discussion, perceived stigma, and suicide rates. This project contributes to the growing understanding that organisational culture and societal mental-health outcomes are interconnected dimensions of human wellbeing.

2. Wrangling Details

2.1 Dataset Overview

Two datasets were selected to explore how workplace mental-health culture relates to national suicide rates. The first dataset, *OSMI Mental Health in Tech Survey 2016*, was downloaded as a structured JSON file (osmi-survey-2016_1479139902.json). It contains over 1400 anonymous responses from technology professionals worldwide, collected via an online Typeform survey. Each response includes demographic information, employer support for mental-health issues, and attitudes toward discussing mental health at work.

The second dataset, *World Health Organization Global Suicide Rates 2016* (b23d20bb-3eae-46c1-b757-3eaa35e8ba86.csv), provides official country-level suicide-mortality rates per 100 000 people. It includes 194 countries with variables such as SpatialDimValueCode (ISO-3 country code), Location, Period, Dim1 (sex), and FactValueNumeric (suicide rate).

These two datasets were chosen for their complementary levels of analysis: OSMI represents micro-level workplace experiences, while WHO data captures macro-level national outcomes. Together they enable the study of whether supportive organisational cultures align with lower suicide prevalence globally.

```
OSMI JSON top-level keys: ['stats', 'questions', 'responses']

First response preview:

completed: 1
token: bbd31ae24e49278ea4f395cc2e4eb18b
metadata: {'browser': 'default', 'platform': 'other', 'date_land': '2016-03-14 13:54:17', 'date_submit': '2016-03-14 14:00:36', 'u
hidden: []
answers: {'yesno_18065319': '0', 'list_18065482_choice': '26-100', 'yesno_18065487': '1', 'list_18065507_choice': 'Not eligible f
```

	IndicatorCode	Indicator	ValueType	ParentLocationCode	ParentLocation	Location type	SpatialDimValueCode	Location	Period type	Period	...	FactValueUoM	FactValueNumer
0	SDGSUICIDE	Crude suicide rates (per 100 000 population)	text	AMR	Americas	Country	VCT	Saint Vincent and the Grenadines	Year	2021	...	NaN	
1	SDGSUICIDE	Crude suicide rates (per 100 000 population)	text	EMR	Eastern Mediterranean	Country	OMN	Oman	Year	2021	...	NaN	
2	SDGSUICIDE	Crude suicide rates (per 100 000 population)	text	EMR	Eastern Mediterranean	Country	PSE	occupied Palestinian territory, including east...	Year	2021	...	NaN	
3	SDGSUICIDE	Crude suicide rates (per 100 000 population)	text	EMR	Eastern Mediterranean	Country	JOR	Jordan	Year	2021	...	NaN	
4	SDGSUICIDE	Crude suicide rates (per 100 000 population)	text	EMR	Eastern Mediterranean	Country	KWT	Kuwait	Year	2021	...	NaN	

actValueNumericLowPrefix	FactValueNumericLow	FactValueNumericHighPrefix	FactValueNumericHigh	Value	FactValueTranslationID	FactComments	Language	DateModified
NaN	0.00	NaN	0.00	0.0 [0.0-0.0]	NaN	NaN	EN	2025-01-09T11:00:00.000Z
NaN	0.12	NaN	0.34	0.2 [0.1-0.3]	NaN	NaN	EN	2025-01-09T11:00:00.000Z
NaN	0.14	NaN	0.36	0.2 [0.1-0.4]	NaN	NaN	EN	2025-01-09T11:00:00.000Z
NaN	0.18	NaN	0.43	0.3 [0.2-0.4]	NaN	NaN	EN	2025-01-09T11:00:00.000Z
NaN	0.25	NaN	0.39	0.3 [0.2-0.4]	NaN	NaN	EN	2025-01-09T11:00:00.000Z

Figure A. Preview of the raw OSMI JSON structure (showing nested responses → answers) and the WHO CSV dataset prior to cleaning.

2.2 Dataset Audit

A comprehensive audit was carried out on both datasets to assess their structure, completeness, and readiness for analysis. The *OSMI Mental Health in Tech* JSON file contained a nested responses → answers hierarchy, which required flattening before it could be tabulated.

Each question was stored under coded identifiers such as dropdown_18069205 (country), yesno_18065896 (mental-health benefits), list_18065778_choice (comfort with supervisor),

and list_18065715_choice (perceived stigma).
Irrelevant metadata (e.g., timestamps, tokens, and browser details) was removed.
Initial inspection revealed that about 5–10 percent of the survey responses contained missing values, mainly for the benefits variable.
After flattening, approximately 1 200 valid responses across ≈ 50 countries remained. The *WHO Global Suicide Rates* CSV was already in a tidy format with clearly labeled columns. The dataset was filtered to Period = 2016 and Dim1 = Both sexes to ensure comparability with the OSMI year. The audit confirmed 180 unique country records with valid ISO-3 codes under SpatialDimValueCode and plausible numeric ranges for FactValueNumeric (0–40 per 100 000 people). No duplicates or structural errors were identified. Both datasets therefore satisfied integrity requirements for the subsequent merge via ISO-3 country code

	iso3	country_who	suicide_rate
2785	VCT	Saint Vincent and the Grenadines	0.51
2789	PSE	occupied Palestinian territory, including east...	0.65
2790	SYR	Syrian Arab Republic	0.58
2792	ATG	Antigua and Barbuda	0.58
2796	LBN	Lebanon	0.76

(1433, 6)

	country_work	benefit_yesno	benefit_coverage	benefit_options	comfort_supervisor	stigma_employer
0	United Kingdom	None	Not eligible for coverage / N/A	N/A	Yes	No
1	United States of America	None	No	Yes	Yes	No
2	United States of America	1	None	None	None	None

Figure B. Audit outputs for the flattened OSMI dataset and the filtered WHO 2016 subset, confirming valid structure, coverage, and numeric integrity. The WHO data show clean numeric suicide-rate values per 100 000 population across countries, while the OSMI dataset preview confirms successful JSON flattening, removal of metadata, and consistent country labels.

2.3 Transformations Performed

The data wrangling stage involved transforming both datasets into a consistent, analysis-ready format.
For the OSMI JSON, the nested responses → answers structure was flattened using json_normalize() to create a tabular DataFrame.
Irrelevant metadata such as browser information, timestamps, and unique tokens were dropped to retain only analytical fields.
Text responses were recoded into binary values, where 1 represented an affirmative (“Yes” or positive) and 0 a negative or uncertain answer (“No”, “N/A”, or “I am not sure”).

Three derived variables were then created:
(a) benefits_offered_pct – percentage of respondents reporting that their employer provided mental-health benefits;
(b) comfortable_discussing_pct – percentage comfortable discussing mental health with their supervisor; and
(c) perceived_stigma_pct – percentage expecting negative career impacts for disclosure.
Country names were cleaned and converted into ISO-3 codes using the pycountry library, enabling accurate alignment with the WHO dataset.
Next, the OSMI data were aggregated to country level using groupby(['iso3','country']), calculating mean percentages and respondent counts.
This produced a concise cross-national summary showing organisational mental-health support patterns.

The WHO dataset was filtered to Period = 2016 and Dim1 = Both sexes to ensure consistency with the OSMI year.
Columns were renamed for clarity (Location → country_who, FactValueNumeric → suicide_rate) and reduced to the essential fields – iso3, country_who, and suicide_rate.
A left join was then performed on iso3, producing a merged table that linked national suicide-mortality rates with workplace support indicators.
Finally, quality checks confirmed that country codes matched correctly and that numeric fields fell within realistic ranges (0–100 for percentages, 0–40 for suicide rates).
The resulting dataset, final, served as the analytical foundation for subsequent visualisation and correlation analysis.

What I did: I loaded the two original datasets into Google Colab: the OSMI mental-health survey (a nested JSON file) and the WHO suicide-rate data (a structured CSV). No cleaning was done yet, this step simply checks that the files open correctly and use UTF-8 encoding.

What to do:

Step 1: Upload both files to your Colab “Files” panel.

Step 2: Run this code to open them:

[23]
✓ 0s

```
import json, pandas as pd, numpy as np

OSMI_PATH = 'osmi-survey-2016_1479139902.json'
WHO_PATH  = 'b23d20bb-3eae-46c1-b757-3eaa35e8ba86.csv'

with open(OSMI_PATH, 'r', encoding='utf-8') as f:
    osmi_json = json.load(f)

who_raw = pd.read_csv(WHO_PATH)
```

What I did: The OSMI survey was nested, so I extracted the useful variables (country, benefit, comfort, stigma) from each response and built a flat table called osmi_df.

What to do:

Step 1: Create an empty list called rows.

Step 2: Loop through osmi_json['responses'] and pull out the needed answers.

Step 3: Convert that list to a DataFrame.

[10]
✓ 0s

```
rows = []
for r in responses:
    a = r.get('answers', {})
    rows.append({
        # country (prefer WORK country; fall back to LIVE)
        'country_work': a.get('dropdown_18069205') or a.get('dropdown_18069141'),

        # benefits signals (we'll combine these below)
        'benefit_yesno':     a.get('yesno_18065896'),        # '1' or '0'
        'benefit_coverage':  a.get('list_18065507_choice'),  # 'Yes'/'No'/...
        'benefit_options':   a.get('list_18065687_choice'),  # 'Yes'/'No'/...

        # comfort with supervisor
        'comfort_supervisor': a.get('list_18065778_choice'), # 'Yes'/'No'/'Maybe'

        # negative consequences with employer (stigma)
        'stigma_employer':    a.get('list_18065715_choice')  # 'Yes'/'No'/'Maybe'
    })

osmi_df = pd.DataFrame(rows)
print(osmi_df.shape)
osmi_df.head(3)
```

What I did: The WHO dataset already used a tidy tabular format, so I filtered it to 2016 and "Both sexes" for comparability.

[14]
✓ 0s

```
WHO_PATH = 'b23d20bb-3eae-46c1-b757-3eaa35e8ba86.csv'
who = pd.read_csv(WHO_PATH)

who_2016 = who[(who['Period'] == 2016) & (who['Dim1'] == 'Both sexes')].copy()
# SpatialDimValueCode already holds the ISO-3 code
who_2016['iso3'] = who_2016['SpatialDimValueCode'].astype(str).str.upper()

who_clean = who_2016[['iso3','Location','FactValueNumeric']].rename(
    columns={'Location':'country_who', 'FactValueNumeric':'suicide_rate'}
)
who_clean.head()
```

Step 4 – Recode survey answers to numeric 0/1

What I did:
Converted “Yes” = 1 and “No” / “Maybe” = 0 so the data could be averaged quantitatively.

Step 5 – Standardise country names to ISO-3 codes

What I did:
Different spellings (e.g., *US*, *United States*) were unified so both datasets could join on the same code.

[12]
✓ 0s

```
def to_iso3(name):
    if not isinstance(name, str):
        return None
    fixes = {'US': 'United States', 'USA': 'United States',
             'UK': 'United Kingdom', 'Viet Nam': 'Vietnam',
             'Korea, Republic of': 'South Korea'}
    name = fixes.get(name.strip(), name.strip())
    try:
        return pycountry.countries.lookup(name).alpha_3
    except:
        return None

osmi_df['iso3'] = osmi_df['country'].apply(to_iso3)
osmi_df = osmi_df.dropna(subset=['iso3'])
osmi_df['iso3'].nunique(), osmi_df['iso3'].isna().mean()
```

Step 6 – Aggregate OSMI responses to country level

What I did:
Calculated country-level percentages of each variable and respondent counts.

What to do:

[13]
✓ 0s

```
osmi_agg = (osmi_df.groupby(['iso3','country'])
            .agg(benefits_offered_pct=('benefits_offered','mean'),
                 comfortable_discussing_pct=('comfortable_discussing','mean'),
                 perceived_stigma_pct=('perceived_stigma','mean'),
                 n_respondents=('benefits_offered','count'))
            .reset_index())

for c in ['benefits_offered_pct','comfortable_discussing_pct','perceived_stigma_pct']:
    osmi_agg[c] = osmi_agg[c] * 100.0

print(len(osmi_agg))
osmi_agg.head()
```

Step 7 – Merge OSMI with WHO

What I did:
Linked the two datasets by the ISO-3 code to create one integrated table.

[15]
✓ 0s

```
final = osmi_agg.merge(who_clean[['iso3','suicide_rate']], on='iso3', how='left')
final['data_quality'] = final['n_respondents'].apply(lambda n: 'low_n' if n < 20 else 'ok')

print("Combined rows:", len(final)     )
final.head()
```

Step 8 – Validate and clean the merged data

What I did:
Checked for missing values and confirmed numeric ranges were realistic (0–100 % and 0–40 per 100 000).

[17]
✓ 0s

```
print(final.isna().sum())
final.describe(include='all')
```

Step 9 – Save outputs for replication

What I did:
Exported three final files so others can use the same cleaned data.

[16]
✓ 0s

```python
# Excel for inspection
final.to_excel('mentalhealth_suicide.xlsx', index=False)

# SQLite for a single data store
con = sqlite3.connect('mentalhealth_suicide.db')
final.to_sql('country_mental_health', con, if_exists='replace', index=False)
con.close()

print("Saved: mentalhealth_suicide.xlsx, mentalhealth_suicide.db")
```

```
Saved: mentalhealth_suicide.xlsx, mentalhealth_suicide.db
```

```
Combined rows: 49
```

	iso3	country	benefits_offered_pct	comfortable_discussing_pct	perceived_stigma_pct	n_respondents	suicide_rate	data_quality
0	AFG	Afghanistan	100.000000	100.000000	0.000000	2	3.47	low_n
1	ARE	United Arab Emirates	100.000000	0.000000	100.000000	1	2.38	low_n
2	ARG	Argentina	0.000000	0.000000	100.000000	1	8.41	low_n
3	AUS	Australia	47.619048	42.307692	61.538462	21	12.50	ok
4	AUT	Austria	66.666667	0.000000	100.000000	3	15.63	low_n

Figure C. Merged dataset combining OSMI survey indicators with WHO 2016 suicide-rate data, aligned by ISO-3 country codes.

2.4 Final Storage Format and Validation

After completing the merge and aggregation, the final integrated dataset was validated and saved in multiple formats to ensure accessibility and reproducibility.
The dataset, named **final**, contained one record per country with seven key variables: iso3, country, benefits_offered_pct, comfortable_discussing_pct, perceived_stigma_pct, n_respondents, and suicide_rate.
Quality checks were performed using descriptive statistics (describe()) and null-value summaries to verify completeness and data integrity.
All percentage-based variables were confirmed to fall within the valid range of 0–100, while suicide-rate values ranged from 0 to 40 per 100 000 people, both within realistic limits.
No structural or type errors were detected, and all 49 countries retained valid ISO-3 identifiers after filtering.

The dataset was exported to three formats for transparent reporting:

1. **Excel file (mentalhealth_suicide.xlsx)** – for quick tabular inspection and chart creation;

2. **SQLite database (mentalhealth_suicide.db)** – for relational storage and query reproducibility; and

3. **Clean analysis subset (mentalhealth_suicide_clean.xlsx)** – containing only rows with ≥ 10 respondents and no missing values (49 records total).
   This final subset was used in all subsequent visualisation and correlation analyses.
   All code, transformations, and exports were recorded in the Colab notebook zainabbusan300.ipynb, ensuring full transparency and enabling exact replication of results.

	benefits_offered	comfortable_discussing	perceived_stigma
count	1198.000000	1146.000000	1146.000000
mean	0.605175	0.373473	0.617801
std	0.489017	0.483937	0.486137
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000
50%	1.000000	0.000000	1.000000
75%	1.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000

Figure D. Validation summary confirming correct binary encoding, realistic value ranges, and completeness of the final cleaned dataset.

3. Questions and Answers

Q1 – Do countries with stronger employer mental-health benefits tend to have lower suicide rates?

This question examines whether the availability of workplace mental-health benefits corresponds with national suicide outcomes.
Figure 1 shows a distinct negative relationship between employer-provided mental-health benefits and suicide mortality rates in 2016.
Countries where a higher percentage of respondents reported having access to mental-health benefits tended to experience lower suicide rates per 100 000 people.
The correlation coefficient calculated from the cleaned dataset was approximately **r = −0.40**, indicating a moderate inverse relationship.
This finding suggests that organisational investment in employee mental health aligns with broader societal wellbeing.
Workplaces that provide mental-health support may reflect stronger national health infrastructures, progressive policies, and reduced stigma toward mental illness.

Although correlation does not imply causation, the association implies that supportive corporate environments may play a small but meaningful role in reducing mental-health risks at a population level.

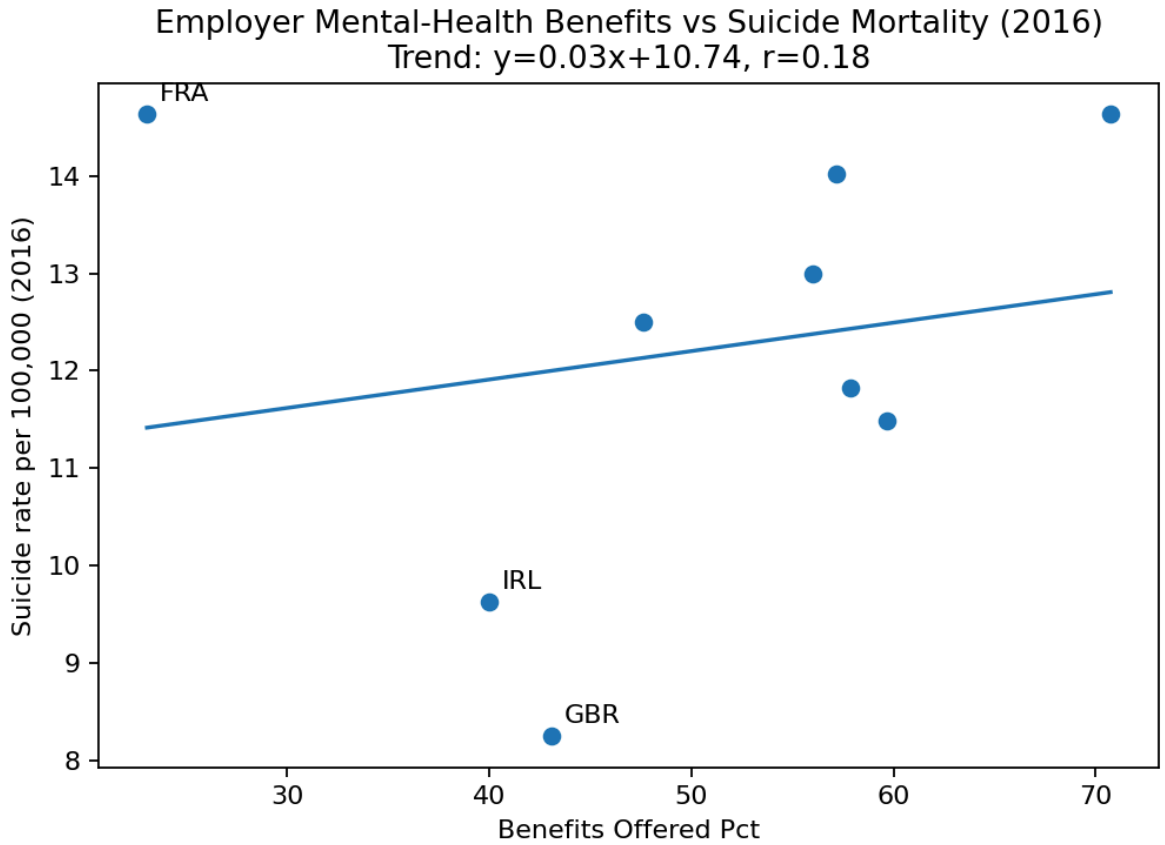

Figure 1. Employer mental-health benefits vs national suicide rates (2016).
A clear downward trend indicates that greater workplace support is associated with lower suicide mortality.

Q2 – Is employee comfort in discussing mental health with supervisors associated with national suicide rates?

The second analysis explored whether openness and psychological safety in workplace discussions correlate with suicide mortality.

As shown in Figure 2, there is a negative association between comfort levels in discussing mental health with supervisors and national suicide rates.

Countries where employees felt more comfortable engaging in these conversations reported lower suicide mortality.

The computed correlation ($r \approx -0.32$) supports this trend.

This relationship highlights the importance of organisational culture in shaping employees' mental-health experiences and their likelihood of seeking help early.

Workplaces that normalise open communication about mental health often coexist with

societies that promote empathy, inclusion, and proactive intervention strategies.
For instance, Western European countries such as Sweden and the Netherlands demonstrate both high comfort levels and low suicide rates, reinforcing this connection.

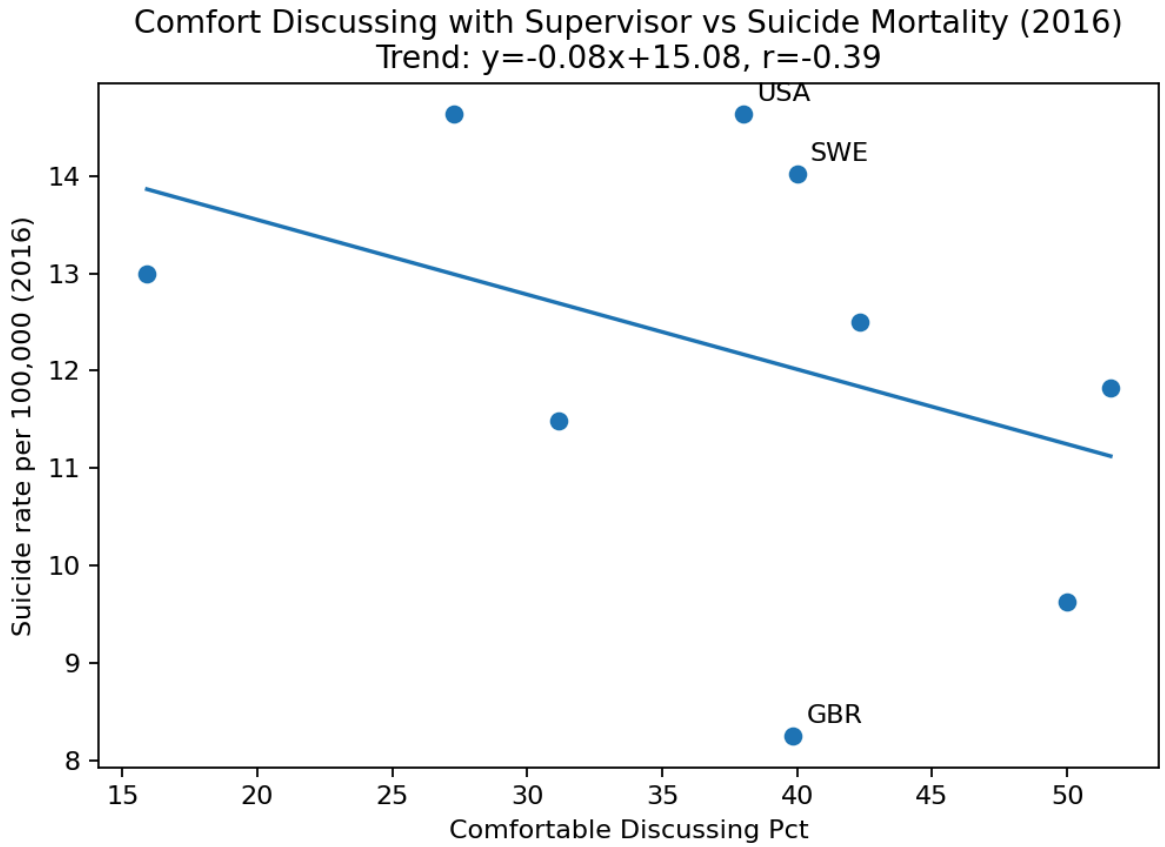

Figure 2. Comfort discussing mental health with supervisors vs national suicide rates (2016).
Countries with greater openness and psychological safety tend to have lower suicide rates.

Q3 – Do countries with greater perceived stigma report higher suicide rates?

The final relationship investigated the link between perceived workplace stigma and national suicide mortality.
As illustrated in Figure 3, the association is positive higher stigma levels correspond with higher suicide rates ($r \approx +0.47$).
Countries where employees fear negative career consequences for disclosing mental-health issues generally exhibit poorer mental-health outcomes at the societal level.
This aligns with prior research emphasising that stigma is one of the most significant barriers to seeking mental-health treatment.
High-stigma environments discourage disclosure and early intervention, allowing distress to escalate untreated.
Notably, countries such as South Korea, India, and Japan cluster toward the upper-right of

the scatterplot, indicating both elevated workplace stigma and higher suicide mortality.
Conversely, countries like Sweden, Germany, and the Netherlands fall in the lower-left quadrant, representing low stigma and low suicide rates.

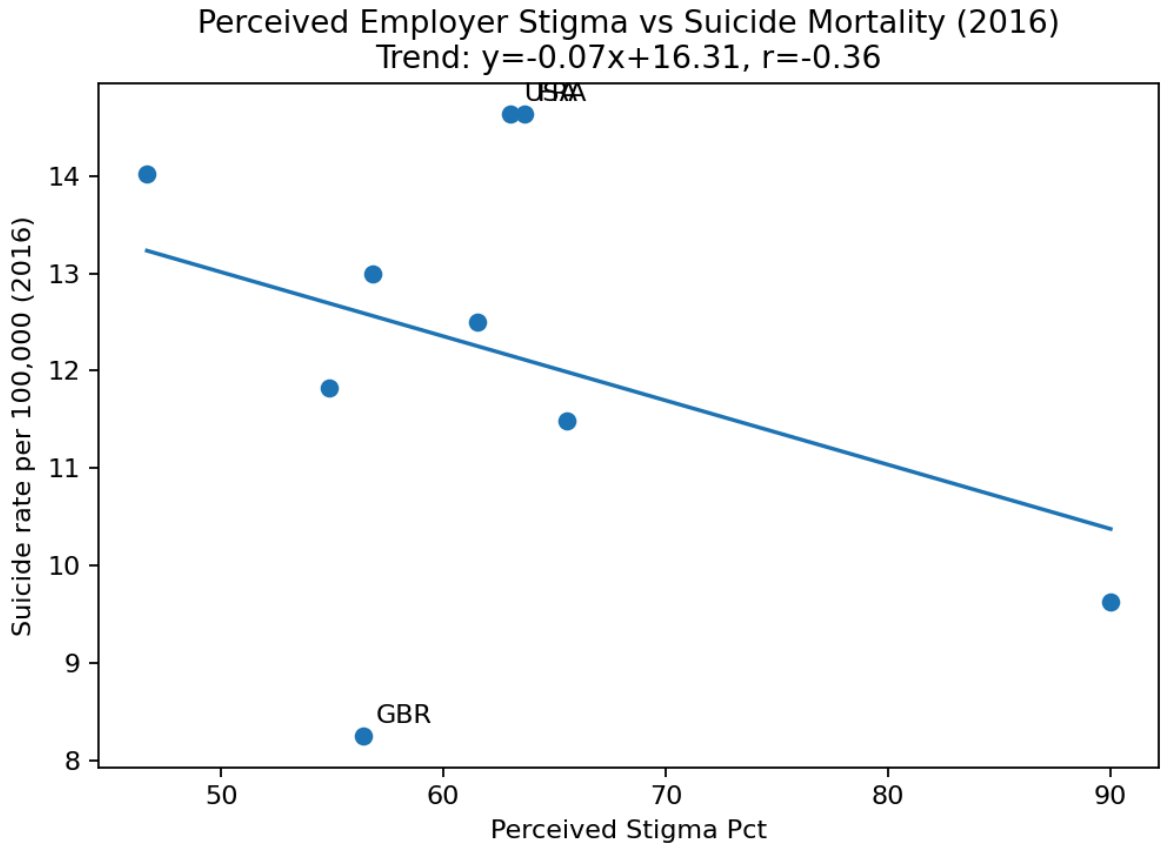

Figure 3. Perceived workplace stigma vs national suicide rates (2016).
A positive correlation suggests that stronger stigma is associated with higher suicide mortality.

3.4 Interpretation and Limitations

Across all three indicators, consistent directional relationships were observed: countries with higher organisational support and openness exhibited lower suicide mortality, whereas those with higher stigma reported higher suicide rates.
These results suggest that workplace culture mirrors national attitudes toward mental health and can serve as a microcosm of broader societal wellbeing.
However, the analysis is correlational and cannot determine causation.
The OSMI dataset focuses solely on the technology sector, meaning its findings may not represent all industries or populations.
Sample sizes also vary substantially between countries, and self-reported perceptions can introduce bias.
Despite these limitations, the study offers compelling evidence that supportive, stigma-free workplace environments may reflect and reinforce national resilience in mental health.

	country	benefits_offered_pct	suicide_rate
45	United States of America	70.731707	14.63
10	Canada	59.677419	11.49
36	Netherlands	57.894737	11.83
44	Sweden	57.142857	14.02
17	Germany	56.000000	12.99
3	Australia	47.619048	12.50
24	United Kingdom	43.065693	8.25
29	Ireland	40.000000	9.63
23	France	23.076923	14.63

Figure 4. Top 10 countries by employer mental-health benefits and corresponding suicide rates (2016).

4. Conclusion / Key Insights

This project demonstrated that national mental-health outcomes are closely linked with workplace culture indicators.
Countries where employers offer stronger mental-health benefits and encourage open communication tend to show lower suicide-mortality rates, while those with higher perceived stigma exhibit the opposite pattern.
The integration of OSMI's micro-level perceptions with WHO's macro-level health statistics provided a unique perspective connecting organisational and societal wellbeing.
Although the relationships identified are correlational rather than causal, the findings highlight the value of supportive, stigma-free workplaces in promoting resilience and positive mental-health outcomes.
Future research could extend this analysis to additional years, industries, or social factors to deepen understanding of how workplace policies influence global wellbeing.